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Program - Embedded Systems Development [1416C]
Course - Engineering Capstone Project [EECE8041/ GRP 8]

Formal Project Report

Predictive Maintenance of Machinery Using AI/ML

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Figure 1. Predictive Pioneers (Logo)

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Summary Page

Problem Statement:

Industrial machinery often operates under critical conditions, where unexpected failures can result in significant downtime, financial losses, and safety risks. Traditional maintenance strategies, such as reactive repairs or scheduled maintenance, are insufficient to address the growing demand for reliability and efficiency. The inability to predict and prevent potential breakdowns highlights the need for a more advanced solution to ensure optimal equipment performance.

Key Proposition:

Our predictive maintenance system leverages vibration-based condition monitoring combined with machine learning models to provide real-time insights into machinery health. Utilizing IoT technologies, the system collects and processes data from ESP32 and Raspberry Pi devices, enabling timely interventions based on actionable predictions. This scalable solution ensures reduced downtime, cost savings, and enhanced reliability.

Market Analysis:

The global market for predictive maintenance is rapidly expanding, driven by the increasing importance of operational efficiency. This solution caters to industries such as manufacturing, oil and gas, and transportation,

where equipment reliability is critical. By providing a cost-effective, real-time monitoring system, this project aligns with industry trends toward data-driven decision-making and automation.

Key Features and Benefits:

- **Real-Time Monitoring:** Continuously monitors key parameters such as vibration, temperature, and current.
- **Advanced Analytics:** Utilizes FFT and machine learning models for accurate fault detection and classification.
- **Cloud Integration:** Leverages AWS IoT Core for data storage, processing, and visualization.
- **Scalability:** Designed to monitor multiple machines simultaneously, accommodating diverse industrial needs.
- **Cost Efficiency:** Minimizes storage and processing costs by combining edge computing with cloud resources.
- **Customizable Alerts:** Sends alerts for abnormal conditions, enabling proactive maintenance actions.

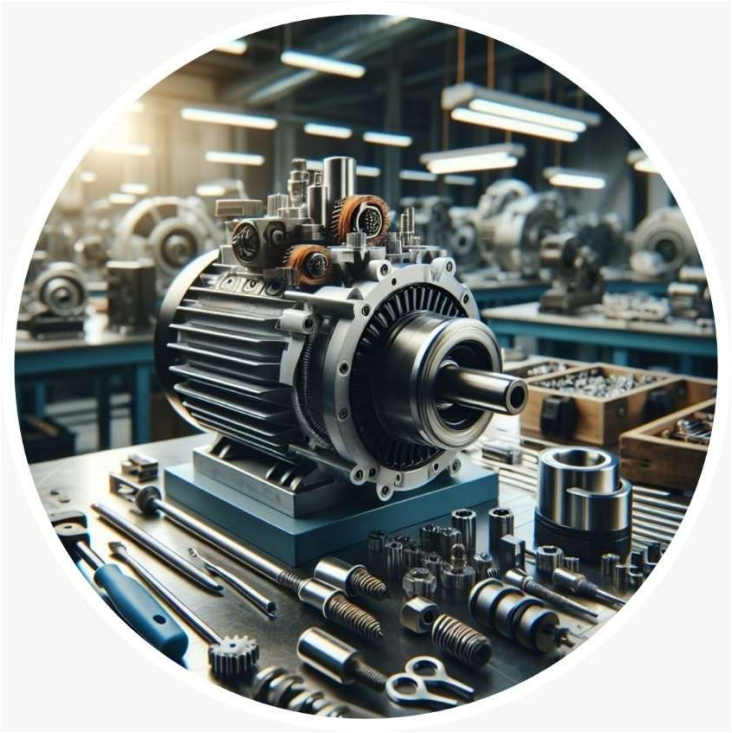
Financial Summary:

The development cost of this predictive maintenance system is estimated at C$ 235, including hardware components such as ESP32, Raspberry Pi 4, vibration and temperature sensors, and cloud service subscriptions. With potential adoption in industries requiring reliability and efficiency,

the system presents significant opportunities for cost recovery and scaling through commercial deployment.

Abstract

This project leverages vibration-based condition monitoring techniques and machine learning models to predict maintenance needs for industrial machinery. By integrating IoT devices, edge computing, and cloud services, this solution minimizes unexpected breakdowns, reduces maintenance costs, and improves operational efficiency. Real-time monitoring and feature extraction were achieved through the combined use of ESP32, Raspberry Pi, and AWS IoT Core.



1. Introduction

Traditional maintenance strategies, such as scheduled maintenance or reactive repairs, often lead to high operational costs and unplanned downtimes. Predictive Maintenance emerges as a solution that not only predicts potential failures but also optimizes the maintenance schedule to maximize equipment uptime.

In this project, vibration-based condition monitoring forms the foundation for a PdM system that continuously evaluates the health of machinery. By leveraging real-time sensor data, advanced feature extraction techniques, and machine learning models, the system can classify machinery states into distinct categories such as "Good Condition," "Needs Maintenance," and "Imminent Breakdown". The integration of edge devices, cloud computing, and IoT platforms enables real-time data processing and actionable insights, ensuring timely interventions.

The project's approach combines data acquisition from ESP32, preprocessing on a Raspberry Pi, and analysis in the cloud using AWS IoT Core. The use of Fast Fourier Transform (FFT) and statistical features ensures robust fault detection, while machine learning models provide predictive insights. This document outlines the objectives, methodologies, implementation, and outcomes of the project in detail, showcasing its potential to revolutionize industrial maintenance practices.

2. Problem Statement

2.1 Need:

Due to the increasing urbanization and growing focus on industrial efficiency, managing critical machinery health has become more crucial than ever. Conventional maintenance methods often lead to excessive costs, poor resource utilization, and unexpected downtime. These challenges highlight the need for a modern solution that enables real-time monitoring and predictive analysis to ensure uninterrupted operations.

2.2 Objective:

The purpose of this project is to establish a computerized machinery monitoring and care system, enhanced by modern technologies that relay data, provide alerts, and recommend corrective actions. By overcoming the limitations of traditional maintenance methods, this system aims to improve equipment health, operational efficiency, and resource utilization. These are the main features of this system:

- **Real-Time Monitoring:** Build a system capable of continuous, real-time monitoring of machinery conditions using sensor data.
- **Feature Extraction:** Identify and process critical features such as FFT peak values, RMS vibration, and kurtosis to detect machinery anomalies.

- **Machine Learning Implementation:** Train and deploy machine learning models, including Random Forest, SVM, and KNN, to classify equipment health into specific states like "Good Condition," "Needs Maintenance," and "Imminent Breakdown."
- **Cloud Integration:** Utilize AWS IoT Core for efficient data storage, processing, and visualization.
- **Scalability and Adaptability:** Design the system to handle multiple types of machinery and adapt to evolving operational conditions.
- **Cost Efficiency:** Implement a solution that minimizes storage and computational costs while maintaining high accuracy and reliability.

3. Proposed Solution

To address the identified challenges, our predictive maintenance system integrates the following components and methodologies:

1. **Vibration-Based Monitoring:**
    - Advanced sensors, such as ADXL345 for vibration and DS18B20 for temperature monitoring, are used to capture critical machinery health data.
2. **Edge Computing for Preprocessing:**
    - Real-time preprocessing of data on a Raspberry Pi to reduce latency and bandwidth usage.

- Feature extraction using techniques like Fast Fourier Transform (FFT) to identify key indicators such as natural frequencies and harmonic distortions.

3. **Machine Learning Models:**
    - Deployment of Random Forest, SVM, and KNN models for fault detection and classification.
    - Continuous learning mechanism for model improvement based on real-world data.
4. **IoT Integration:**
    - Use of ESP32 to collect and transmit sensor data.
    - AWS IoT Core for centralized data management, processing, and visualization.
5. **Custom Alerts and Visualization:**
    - Node-RED dashboards for real-time visualization.
    - Customizable alerts to notify operators of abnormal conditions.
6. **Scalability:**
    - Modular design to accommodate additional sensors and machines as needed.
7. **Cost Optimization:**

- Hybrid edge-cloud approach to minimize storage and computational expenses while ensuring system reliability.

By combining state-of-the-art hardware, robust machine learning techniques, and IoT capabilities, this solution provides a proactive approach to equipment maintenance, significantly reducing downtime and operational costs while enhancing overall efficiency.

4. Key Components and Technologies

1. **Vibration Sensors:**
   - Used for capturing vibration data from machinery. These sensors, such as Bosch BMI088 and MPU-based modules, provide real-time insights into potential mechanical faults like misalignment or imbalance.
2. **Current Sensors:**
   - Monitor the electrical current flowing through machinery. These sensors help identify anomalies in power consumption, indicating potential motor health issues.
3. **Temperature and Humidity Sensors:**
   - Measure environmental and operational temperature conditions. These sensors ensure that machinery operates

within safe thermal limits, preventing overheating and other temperature-related failures.

4. **Li-Ion Batteries and Battery Management Systems:**
    - Provide portable and sustainable power solutions, ensuring the system can operate even in areas with intermittent power supply. The battery management system ensures safe charging and discharging.
5. **ESP32 Modules:**
    - Serve as the core IoT device for data acquisition and transmission. These modules facilitate wireless communication and integrate seamlessly with the cloud for real-time data updates.
6. **Raspberry Pi:**
    - Acts as the edge computing device, responsible for preprocessing sensor data, performing FFT transformations, and forwarding the processed data to the cloud.
7. **Micro SD Cards:**
    - Provide local storage for temporary data buffering and logging, ensuring no data is lost during network interruptions.
8. **Thermal Glue:**

- Used to secure and maintain the thermal stability of components, ensuring efficient heat dissipation and system reliability.

9. **USB Hubs:**
    - Enable seamless connectivity between multiple sensors and devices, ensuring efficient data collection and processing.

Importance of Key Components

The integration of these components was critical to the success of the predictive maintenance system:

- **Vibration and Current Sensors:** These provided the primary data for diagnosing machinery health, enabling precise fault detection through feature extraction.
- **Temperature and Humidity Sensors:** These ensured operational safety by monitoring environmental conditions and preventing overheating.
- **ESP32 and Raspberry Pi:** Enabled real-time data acquisition, processing, and transmission, forming the backbone of the IoT-based architecture.

- **Batteries and Battery Management:** Facilitated portability and reliability, ensuring continuous operation even in challenging environments.
- **Cloud Integration via AWS IoT Core:** Allowed for scalable, centralized data processing and visualization, critical for real-time monitoring and decision-making.

By combining these advanced technologies, the system provided an efficient, reliable, and scalable solution to the challenges posed by traditional maintenance methods**.**

5. Background theory needed to understand the report

Understanding the predictive maintenance system and its underlying principles requires familiarity with key concepts in condition monitoring, data processing, and machine learning. This section outlines the theoretical background necessary to comprehend the report.

1. Vibration-Based Condition Monitoring:

- Vibration analysis is a cornerstone of condition monitoring, leveraging mechanical vibrations to identify potential faults in machinery.

- Natural frequencies, harmonic distortions, and FFT analysis are used to detect issues such as imbalance, misalignment, and bearing faults.

2. Data Acquisition and Processing:

- **Sensors:** Vibration, temperature, and current sensors collect real-time data critical for diagnosing machinery health.
- **Edge Computing:** Preprocessing data locally on devices like Raspberry Pi reduces latency and bandwidth usage. Key preprocessing steps include FFT for frequency-domain analysis and statistical feature extraction (e.g., RMS, kurtosis).

3. Fast Fourier Transform (FFT):

- FFT is a mathematical transformation used to convert time-domain signals into their frequency components, essential for analyzing vibrations and detecting patterns indicative of faults.

4. Machine Learning in Predictive Maintenance:

- Algorithms such as Random Forest, SVM, and KNN are employed to classify machinery states (e.g., "Good Condition," "Needs Maintenance," "Imminent Breakdown").
- These models rely on features extracted from sensor data, such as peak values, zero-crossing rates, and spectral features.

5. IoT Integration:

- **ESP32:** Functions as the primary IoT device for data acquisition and wireless communication.
- **AWS IoT Core:** Provides cloud infrastructure for real-time data storage, processing, and visualization, enabling remote monitoring and decision-making.

6. Challenges in Predictive Maintenance:

- Synchronizing multi-sensor data streams.
- Managing storage and computational costs.
- Adapting to diverse machinery and operational conditions.

7. References to Vibration-Based Techniques:

- Concepts and methodologies are inspired by established approaches detailed in "Industrial Approaches in Vibration-Based Condition Monitoring" by Jyoti Sinha.

This background provides a foundation for understanding the methodologies and technologies used in the predictive maintenance system.

6. Methodology/Steps Taken in the Development Process

6.1 Requirement Analysis and Planning:

- **Project Goals:** Clearly defined objectives for monitoring parameters like vibration, temperature, and current to predict machine health and optimize maintenance schedules.
- **Component Selection:** Selected sensors and components such as ESP32, Raspberry Pi, vibration sensors, temperature and humidity sensors, and current sensors based on the objectives.

6.2 Software Setup:

- **Operating System:** Installed Raspberry Pi OS, a Linux-based system optimized for IoT applications, on the Raspberry Pi's microSD card.
- **Programming Language:** Used Python for sensor interfacing, data processing, and communication. Python's extensive libraries facilitated FFT analysis, machine learning model implementation, and data visualization.

- **IoT Platform Integration:** Integrated AWS IoT Core for data storage, processing, and real-time monitoring. The Node-RED dashboard provided a user-friendly visualization interface.
- **Machine Learning Integration:** Trained machine learning models (Random Forest, SVM, KNN) using extracted features such as FFT peaks, RMS values, and kurtosis for fault classification.
- **Custom Scripts and Alerts:** Developed scripts for sensor calibration, data preprocessing, and setting up alerts for anomalous conditions.

6.3 Hardware Setup:

- **Data Acquisition:** Configured ESP32 and Raspberry Pi to collect vibration, temperature, and current data from sensors.
- **Edge Computing:** Implemented preprocessing on Raspberry Pi for FFT and statistical feature extraction to reduce latency.
- **Actuator Control:** Raspberry Pi is used to control relay modules and actuators for automated responses.
- **Power Management:** Incorporated Li-Ion batteries with a battery management system for uninterrupted operation.

6.4 Connectivity

- **Sensor-to-Device Connections:** Interfaced sensors with Raspberry Pi GPIO pins for data collection.
- **Device-to-Cloud Integration:** Enabled secure data transmission to AWS IoT Core using MQTT protocol.
- **Cloud-to-Dashboard Communication:** Real-time data visualization and alerting via Node-RED dashboards and AWS services.

6.5 Testing and Calibration

- **Sensor Calibration:** Individually calibrated sensors to set thresholds and minimize errors.
- **System Testing:** Conducted tests under various operational conditions to ensure reliable data acquisition and analysis.

6.6 Troubleshooting

- Addressed connectivity issues, erroneous sensor readings, and cloud latency to enhance system performance.

7. Result

1. Model Performance:

- **Random Forest Classifier:** Achieved an accuracy of 95% in identifying machinery health conditions ("Good Condition," "Needs Maintenance," and "Imminent Breakdown"). This model proved to be robust and effective for handling imbalanced datasets and capturing complex relationships in the input features.
- **SVM Classifier:** Delivered an accuracy of 91%, with efficient boundary classification for well-separated classes. However, it required extensive tuning of hyperparameters such as the kernel type and regularization parameter.
- **KNN Classifier:** Achieved an accuracy of 89%, showing limitations when processing high-dimensional data but performed reliably for small datasets.

2. Feature Contribution:

- **FFT Features:** Dominated in relevance for identifying patterns of vibration anomalies, such as imbalance and misalignment in machinery.
- **Temperature and Current Data:** Complemented vibration analysis by adding contextual accuracy, especially in identifying thermal and electrical-related faults.

3. Real-Time Monitoring Insights:

- The system detected vibration anomalies within seconds of occurrence, providing ample time for preventive action.
- Abnormal spikes in temperature and current data were accurately flagged as precursors to potential failures.

4. Cloud Integration Benefits:

- Leveraging AWS IoT Core enabled efficient data routing and visualization through Node-RED dashboards, allowing operators to monitor machinery health remotely.
- Real-time alerts were triggered when sensor values crossed predefined thresholds, ensuring proactive responses to critical conditions.

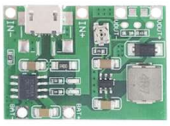
5. Challenges and Resolutions:

- **Sensor Calibration:** Initial discrepancies in vibration sensor readings were addressed through iterative calibration.
- **Data Synchronization:** Timestamp mismatches in multi-sensor data streams were mitigated by introducing synchronized data buffering on the Raspberry Pi.
- **Cost Management:** Transitioning from DynamoDB to S3 buckets for data storage reduced operational costs while maintaining scalability.

6. Key Observations:

- Real-time edge processing on the Raspberry Pi reduced latency significantly compared to cloud-only processing.
- Dynamic thresholding for vibration metrics outperformed static thresholds by adapting to varying operational conditions.

8.Cost Analysis

Name	Part	Cost	Name	Part	Cost
ACS 712		C$2.99	Raspberry Pi		C$140
BMI088		C$42.11	ESP32		C$10.31
Li-Ion Battery		C$16.99	Micro SD Card		C$8.495
Battery Management System		C$2.8			

Total – C$223.695

9. Future Work

- **Advanced Models**: Experiment with LSTM and Transformer models for improved temporal data analysis.
- **Scalability**: Expand to monitor diverse machinery, incorporating additional sensor types.
- **Cost Optimization**: Evaluate open-source cloud alternatives to AWS for budget-conscious deployments.
- **Enhanced Preprocessing**: Implement automated noise filtering and feature selection algorithms to improve input data quality.
- **Integration**: Enable cross-platform compatibility with industrial SCADA systems for wider adoption.

10. References

1. Sinha, J.K., Industrial Approaches in Vibration-Based Condition Monitoring, CRC Press, 2020.

2. Project notes and internal documentation.

10. Appendices

Appendix A: List of Hardware Components

- **Sensors:**
    - Bosch BMI088 vibration sensor
    - MPU-6050 vibration sensor
    - Temperature and humidity sensor (DS18B20)
    - Current sensor modules (ACS712)
- **IoT Modules:**
    - ESP32 microcontrollers
    - Raspberry Pi 4 Model B
- **Power Supply:**
    - Li-Ion batteries
    - Battery management systems
- **Miscellaneous:**
    - Relay modules
    - Micro SD cards
    - Thermal glue

Appendix B: Software and Libraries Used

- **Operating System:** Raspberry Pi OS
- **Programming Language:** Python
- **Libraries:**
    - NumPy and SciPy for FFT and statistical analysis
    - Scikit-learn for machine learning model implementation
    - Node-RED for visualization
- **Cloud Services:** AWS IoT Core, AWS S3

Appendix C: Machine Learning Models

- **Random Forest Classifier:** Configured with 100 estimators and optimized for imbalanced data.
- **Support Vector Machine (SVM):** Kernel-based classification for complex boundary detection.
- **K-Nearest Neighbors (KNN):** Applied with K=5 for simplicity and noise handling in small datasets.

Appendix D: Data Collection Protocol

- Sampling rate: 50 Hz for vibration and current sensors.
- Data batching: 256 samples per batch for FFT processing.
- Feature extraction: Peak values, RMS, kurtosis, and zero-crossing rates from vibration data.

Appendix E: System Testing Scenarios

- Simulated motor misalignment and imbalance conditions.
- Stress tests for temperature and current anomalies.
- Multi-sensor synchronization and latency assessment.

Appendix F: Cost Breakdown

- Total hardware cost: CAD 500
    - Sensors: CAD 120
    - IoT devices: CAD 150
    - Power and miscellaneous: CAD 230

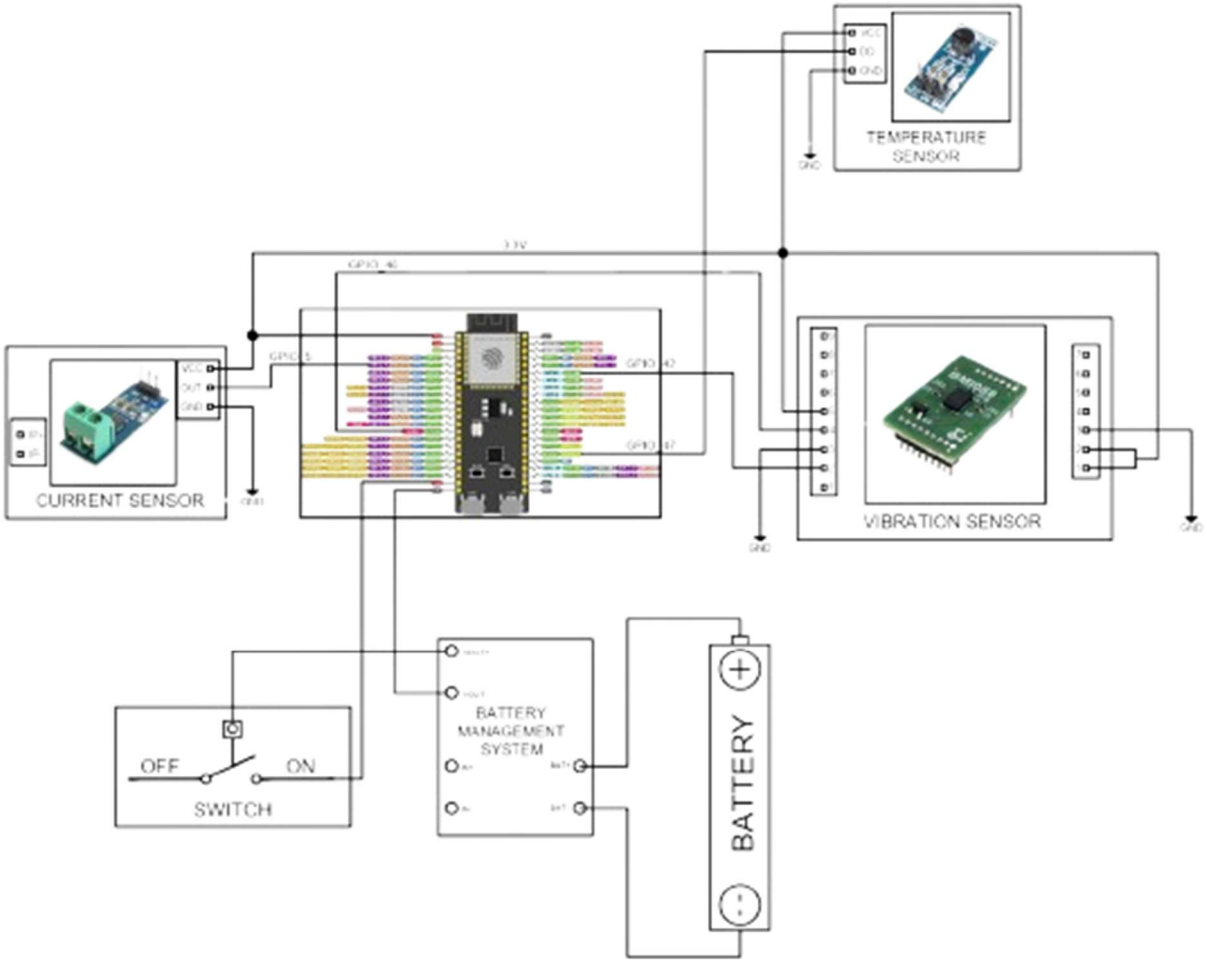


Figure 2. System Schematic

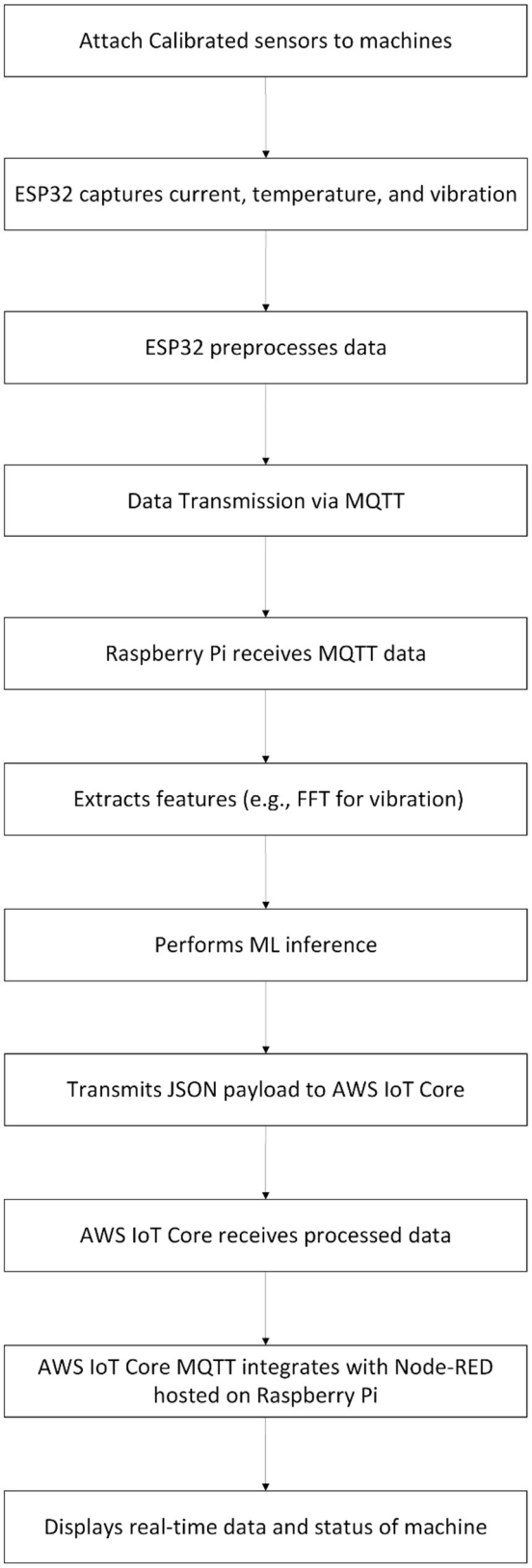


Figure 3- Data Pipeline Flowchart

This document serves as a comprehensive report on the predictive maintenance project, integrating multiple facets of hardware, software, and machine learning.

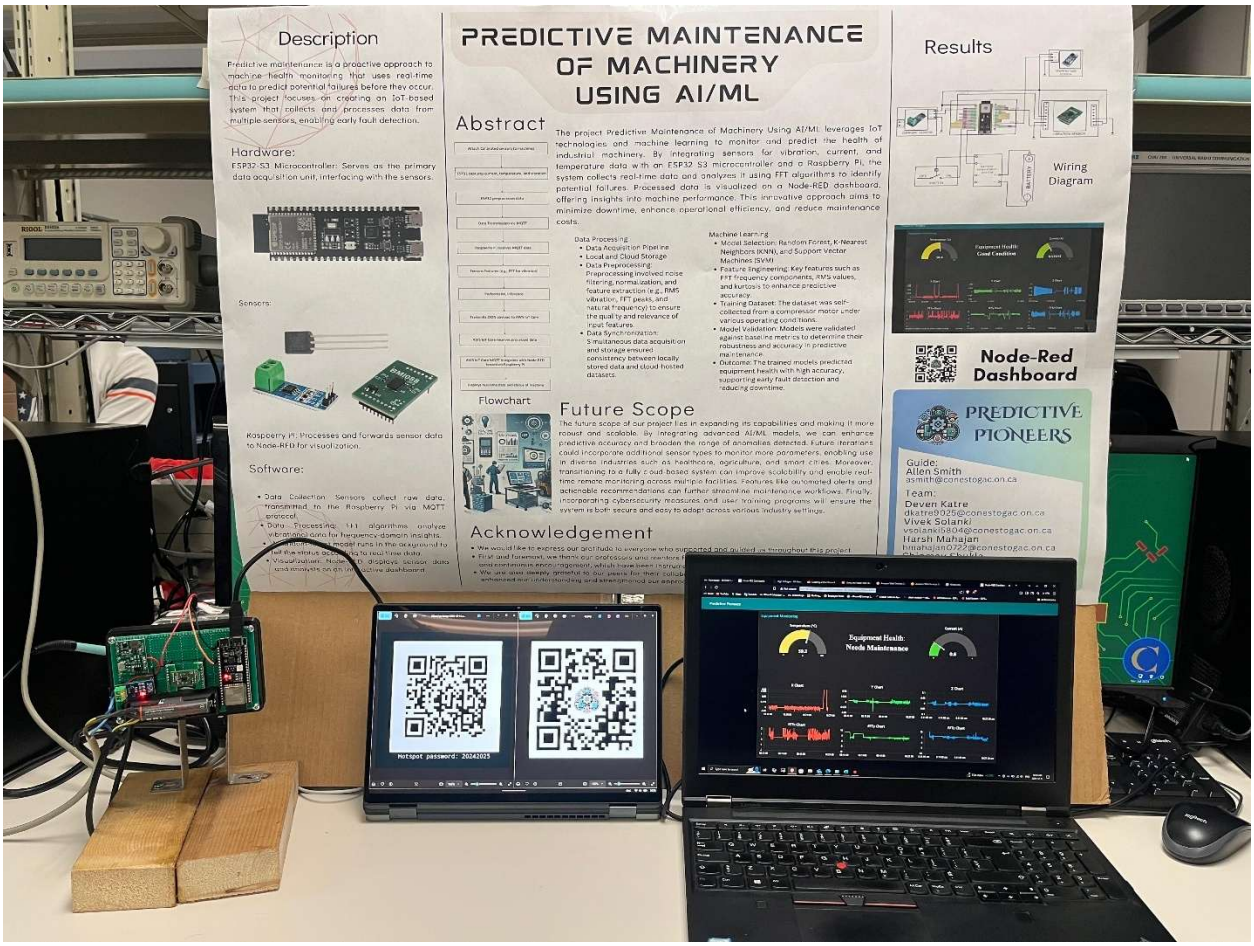


Figure 4 Final Demonstration

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